ANTHONY LIAO

Software Engineer
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SKILLS

Languages: C# · Python · R · SQL · JavaScript · Java · C++ · Mandarin
Tech Stack: .NET · ASP.NET Core · Blazor · WPF · Azure DevOps · SQL Server · Dapper · Git · xUnit
Tools & Data: Visual Studio · Pandas · NumPy · Matplotlib · CatBoost · SHAP · Power BI · Tableau · OpenCV

EXPERIENCE

Software Engineer · **NIAR – Advanced Materials Research** — **Aug 2024 – Present | Wichita, KS**

- Co-led development of PATH, an enterprise web application that replaced a legacy Access database with a modern system for material tracking, machine scheduling, and quality control.
- Designed the backend architecture and wrote recursive SQL queries to trace the full history of how raw materials were blended, reused, and distributed across manufacturing builds — giving engineers complete material genealogy at a glance.
- Engineered ACID-compliant database transactions ensuring inventory updates, build logs, and material recovery either completed fully or rolled back cleanly, preventing any partial or corrupted saves.
- Built a defect reporting system where flagging a build automatically flags all its child parts, and a maintenance scheduler that tracks when each machine is due for service based on time or usage.
- Built frontend workflows including a visual material lineage graph and an Excel-style data grid using Handsontable and JS Interop for high-volume part entry directly in the browser.
- Architected MARS, a desktop tool suite where new analysis modules can be plugged in independently — automating the Excel-heavy data reduction work that engineers previously did by hand after every test.

Software Engineer · **NIAR – ATLAS** — **Aug 2021 – Aug 2024 | Wichita, KS**

- Developed a C# / .NET manufacturing system spanning data acquisition, defect tracking, and WPF interface improvements — integrated with an OpenCV image analysis engine and SQL Server for inspection reporting.
- Integrated an ML-based defect detection system into the AFP manufacturing app, managing the camera data pipeline and surfacing real-time detection accuracy metrics through the WPF interface.
- Improved and maintained an OpenCV-based monitoring system for an injection molding machine to track depth, temperature, and alignment in real time with live accuracy display and run history logging.
- Queried and analyzed large inspection datasets from SQL Server to identify defect patterns and trends in support of engineering reporting.

RESEARCH

Data Science Collaborator · **Independent** — **2025 – Present | Wichita, KS**

- Partnered with medical researchers on statistical analysis, data visualization, and predictive modeling — one paper published and one under review.

PROJECTS

Predicting Unmet Healthcare Needs Among Adults with Disabilities */ R · CatBoost · SHAP · targets*

- Built a survey-weighted machine learning pipeline on CDC BRFSS data (3.9M adults, 2016-2024) to predict unmet healthcare needs among adults with disabilities — comparing CatBoost, Random Forest, and logistic regression with SHAP explainability and subgroup fairness audits across race, income, and disability type.

QUEST: Quiz Understanding & Evaluation for Study Tracking */ C# · WPF*

- Independently built a desktop app that analyzes exported quiz results, identifies weak subject areas, and gives actionable study recommendations.

EDUCATION

M.S. in Data Science — **Wichita State University · May 2026**
Relevant Coursework: Machine Learning, Data Visualization, Artificial Intelligence, Advanced Data Storage.

B.S. in Computer Science · Magna Cum Laude · 3.7 GPA — **Wichita State University · May 2024**
Relevant Coursework: Data Structures, Algorithms, Database Systems, Software Engineering, Operating Systems.